

ME-Brain-1.0: Memory, Cognition and Action for Evolving Embodied Intelligence

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September 2026

Abstract

Current embodied systems largely rely on pretrained capabilities that remain fixed after deployment, limiting their ability to learn from physical interaction. We introduce **MachEmbodied-Brain (ME-Brain)**, a self-evolving embodied system organized around a closed loop of action execution, experience acquisition, experience evolution, and improved execution. **Evolvable Memory** consolidates multimodal trajectories into hierarchical, reusable experience; **Cognitive Core** transforms physical experience into transferable skills; and the **Action Model** combines event-driven keyframes, EventCell local-world prediction, and action-conditioned memory modulation to focus computation on decision-critical moments, regions, and historical evidence. Together, these modules shift embodied intelligence from *train-and-freeze* to *deploy-and-evolve* without model retraining.

Cognitive Core outperforms the strongest comparison models by **8.2** and **9.6 points** on embodied and agent benchmarks. The Action Model achieves **47.88%** mean success on RoboMME, a **3.26-point** improvement over the strongest baseline. On RoboDojo, it reaches a **21.51** mean Score and **16.03%** success rate, exceeding $\pi_{0.5}$ by **10.10** and **9.12 points**. On the six-task ME-RealBench, ME-Brain achieves a **69.5** mean Score and **66.7%** success rate, outperforming DM0.5 by **12.8** and **11.7 points**, respectively.

Project Page: <https://machembodied.com/ME-Brain/ME-Brain-1.0.html>

Code Repository: <https://github.com/MachEmbodied/ME-Brain-1.0>

Contents

1	Introduction	3
2	Related Work	5
2.1	Multimodal Agents	5
2.2	Embodied Vision-Language Models	6
2.3	Embodied Operating Systems	6
3	MachEmbodied-Brain Framework	7
4	Evolvable Memory	8
4.1	Memory Storage and Evolution	9
4.1.1	Multimodal Embodied Memory Storage	9
4.1.2	Memory Evolution	10
4.2	Edge Deployment, Privacy, and Security	11
5	Cognitive Core	11
5.1	Unified Capabilities	11
5.1.1	Embodied Cognitive Capability	11
5.1.2	Multimodal Agentic Capability	13
5.2	Collaboration with Other Modules	13
6	Action Model	14
6.1	Event-Centric World-Action Modeling	14
6.2	Action-Conditioned Memory Modulation	17
7	Experiments	19
7.1	Evaluation of Cognitive Core	19
7.1.1	Embodied Benchmark	19
7.1.2	Agent Benchmark	20
7.2	Evaluation of Action Model	20
7.2.1	Memory Benchmark	21
7.2.2	Action Benchmark	21
7.3	Real-World Robot Evaluation	22
8	Qualitative Examples	24
9	Conclusion and Future Work	30
9.1	Conclusion	30
9.2	Future Work	30
10	Contributions and Acknowledgments	36

1 Introduction

Recent advances in large vision-language models (VLMs) and vision-language-action (VLA) models have substantially improved the perception and control capabilities of embodied systems [1–4]. Embodied brains built on agentic architectures have further enhanced manipulation performance in controlled environments [5–7]. However, existing embodied brains largely assemble the capabilities of pretrained agentic foundation models, with their perception, planning, and action control capabilities remaining largely fixed once model parameters are frozen after training. When confronted with novel environments, objects, or tasks outside the training distribution, these systems must rely on their previously acquired generalization capabilities, rather than learning and improving through continual interaction with the physical world. This fixed-capability paradigm has two fundamental limitations:

- First, a single large-scale training process cannot anticipate all deployment scenarios, making distribution shifts between training data and real-world operating environments difficult to eliminate.
- Second, embodied systems should be capable of continual learning and evolution. Robots accumulate substantial operational experience during long-term deployment, including successful executions, failure diagnoses, and environment-specific adaptation strategies. Yet this experience cannot be incorporated back into the models themselves, leaving valuable learning signals underutilized.

Overcoming these limitations requires systems to continually self-evolve after deployment, an objective that recent embodied-agent systems have begun to explore through experience reuse, skill reflection, and modular adaptation [6, 8, 9]. This raises three interdependent questions: how experience is acquired, how experience evolves, and how experience guides action execution. We introduce ME-Brain, an embodied system comprising three core modules—Evolvable Memory, Cognitive Core, and Action Model—that address these questions, respectively. Together, they establish a continual self-evolution loop of action execution $\rightarrow$ experience acquisition $\rightarrow$ experience evolution $\rightarrow$ action execution, enabling robots to autonomously improve through physical interaction without model retraining.

As the experience hub of this loop, Evolvable Memory addresses the fundamental question of where experience comes from. During each task execution, it structurally encodes multimodal streams, including visual observations, actions, and task states, into task-node-level memory summaries grounded in the execution process. It then constructs an external memory graph, with each complete task represented as a parent node and its subtasks and key events represented as linked child nodes. Maintained independently of the model’s context window, this graph overcomes the constraints of static, length-limited memory in conventional approaches [10–12]. Building on this representation, the system draws inspiration from human memory to establish a three-tier hierarchy of short-, mid-, and long-term memory, enabling continual experience accumulation through a recording–consolidation–abstraction process. Recent work likewise demonstrates that memory at multiple temporal scales and abstraction levels is essential for long-horizon robot control [13, 14]. Short-term memory records temporally dense multimodal state information during task execution, providing immediate historical context for local action decisions. As the task progresses, the system identifies and extracts state transitions, action outcomes, and anomalous feedback that are critical to task progress, organizing them into salient execution records in mid-term memory to provide task-level context for Cognitive Core. Through cross-task event comparison and outcome analysis, the system further distills consistent success conditions, recurring failure modes, and their applicable contexts from mid-term memory into abstract long-term knowledge, supporting retrieval, transfer, and reuse in subsequent tasks.

Cognitive Core serves as the cognitive engine of experience evolution, addressing how cognition continually drives the refinement of accumulated experience. The central challenge is to transform execution processes in the physical world into reusable behavioral capabilities. This requires both accurate representation and understanding of real-world execution states and action–outcome causality, as well as the ability to consolidate and organize experience into skills and strategies, a direction increasingly explored through hierarchical orchestration, execution feedback, and skill-aware reflection [5, 6, 8, 15]. To this end, Cognitive Core integrates embodied cognition with multimodal agent capabilities. Embodied cognition ensures the quality and interpretability of acquired experience through four core capabilities: embodied scene understanding jointly models physical entities and spatial geometry, enabling memory records to represent real-world physical states; long-horizon planning decomposes complex tasks into executable action sequences, allowing experience to be organized around subtask structures; execution understanding provides a causal basis for experience consolidation by interpreting action-induced state transitions; and diagnosis and replanning support execution recovery while continually writing both successful and failed experiences into Evolvable Memory. Multimodal agent capabilities, in turn, provide the mechanisms for transforming experience into reusable competencies. Multimodal understanding determines the information richness and semantic fidelity of experience representations, influencing subsequent retrieval and cross-task analogy. Agent skill invocation encapsulates executable behaviors as Embodied Skills, while a skill-update protocol consolidates effective strategies into new skills, directly enabling experience-to-skill evolution. Long-horizon task understanding maintains state continuity across execution steps and tasks, retrieves relevant experience, and supports dynamic replanning, making it essential for task-level capability transfer. Finally, reasoning and attribution identify the root causes of execution outcomes and distill actionable experience, determining the quality and potential of experience evolution.

Action Model generates executable robot actions from language instructions and serves as the low-level control component through which behavioral improvements are realized. Contemporary generalist policies formulate this mapping with autoregressive or flow-based VLA architectures [2–4], while recent world–action models introduce future visual dynamics as an additional source of action supervision [16, 17]. Motivated by the observation that decision-relevant information in robotic manipulation is concentrated at a small number of critical moments, interaction regions, and historical states, we propose Focus-VLWA, an interaction-event-centric model that augments VLA with a world-model expert to jointly model action generation and event-relevant future prediction. The model organizes action keyframes around salient joint movements and gripper-state changes, reducing the learning overhead associated with stationary and repetitive states. Through EventCell, it focuses on future changes within the interaction region and jointly predicts the target pose of the next grasping or placing event, providing local dynamics supervision and event-level goal guidance for action generation. Through action-conditioned memory modulation, it selectively retrieves historical evidence and directly modulates the action expert, avoiding continual expansion of the vision-language context, in contrast to approaches that retain history through explicit memory tokens or multi-scale context [12, 14, 18]. Together, these designs unify critical moments, interaction regions, and historical evidence within an action-oriented selective modeling framework, concentrating computation on information that directly affects manipulation decisions.

The three modules form a complete, self-driven loop of action execution → experience acquisition → experience evolution → action execution. Each task execution by Action Model produces multimodal trajectories, which Evolvable Memory consolidates into structured experience across its short-, mid-, and long-term hierarchy, transforming both successful execution paths and failure diagnoses into high-quality experience. Grounded in this experience, Cognitive Core uses reasoning and attribution to identify failure causes, multimodal understanding to interpret execution semantics, and long-context management to retrieve historical patterns across tasks. Through its skill invocation mechanism, it

drives experience evolution by consolidating repeatedly validated solutions into new skills and translating long-term experiential knowledge into improved planning strategies. The evolved skill library and memory knowledge base then guide Action Model, enabling improved action trajectories in subsequent tasks and initiating another cycle of experience accumulation and capability refinement. In this way, action execution itself becomes a continual source of learning for ME-Brain.

Our contributions are summarized as follows:

- We introduce ME-Brain, a continually self-evolving embodied system that establishes a complete, self-driven loop of action execution $\rightarrow$ experience acquisition $\rightarrow$ experience evolution $\rightarrow$ action execution, enabling robots to autonomously improve through physical interaction without model retraining.
- We propose Evolvable Memory, which constructs an external memory graph independent of the model’s context window and organizes experience into a human-memory-inspired hierarchy of short-, mid-, and long-term storage. Through a recording–consolidation–abstraction process, it enables continual accumulation and cross-task reuse of multimodal experience, overcoming the constraints of static, length-limited memory in conventional approaches.
- We propose Cognitive Core, which unifies embodied cognition and multimodal cognition. The former grounds experience in accurate and interpretable representations of the physical world through embodied scene understanding, long-horizon planning, execution understanding, and diagnosis and replanning. The latter supports continual experience-to-skill evolution through multimodal understanding, agent skill invocation, long-horizon task understanding, and reasoning and attribution.
- We propose an interaction-event-centric Action Model that augments VLA with a world-model expert to jointly model actions, local future changes, and event target poses. Through keyframe representations, EventCell-based local prediction, and action-conditioned memory modulation, it focuses on critical moments, interaction regions, and historical evidence, reducing computational redundancy while providing historical support and event-level goal guidance for action generation.

2 Related Work

2.1 Multimodal Agents

Recent multimodal agent models increasingly unify visual-language alignment, long-context management, reasoning, and tool use. The Qwen family has progressively strengthened these capabilities from Qwen2.5-VL to Qwen3.8. GLM-5 combines agentic interaction, reasoning, and coding within a single mixture-of-experts architecture and uses asynchronous reinforcement-learning infrastructure and agent RL to improve long-horizon interaction, tool use, and self-correction. Kimi K3 scales this paradigm with a 2.8-trillion-parameter MoE architecture and a million-token context window, applying reinforcement learning across long-horizon coding, general agents, and multimodal reasoning. These general-purpose agents are primarily trained for interaction in digital spaces such as text, code, and web pages. They do not systematically model the spatial structure of the physical world, robot action semantics, or execution-state monitoring. This limitation motivates the explicit integration of multimodal agent capabilities with embodied cognition in ME-Brain.

2.2 Embodied Vision-Language Models

As VLMs have advanced, recent work has transferred general multimodal understanding to physical AI. RynnBrain supports egocentric understanding, spatiotemporal localization, physically grounded reasoning, and perception-aware planning in a unified embodied foundation model. RynnBrain 1.1 adds contact-point prediction, native 3D spatial perception, and a unified cross-embodiment action space. Embodied-R1.5 integrates spatial cognition, task planning and correction, and embodied pointing through a Planner–Grounder–Corrector loop. VeBrain-1.5 reformulates robot control as MLLM-compatible 2D visual-space tasks and unifies perception, spatial reasoning, and control through a shared decision interface [19]. Hy-Embodied-VLM-1.0 organizes physical reasoning around action-related state understanding, action-transition reasoning, and sequence-adaptive reasoning. ACE-Brain-0.5 further couples spatial perception, decision making, embodied interaction, self-monitoring, and self-evolution in a single model [7].

Generalist embodied reasoning models also explore multimodal agent capability and cross-domain transfer. Vesta unifies localization, navigation, embodied question answering, and long-horizon planning with multimodal memory, while MiMo-Embodied jointly trains on autonomous driving and indoor embodied AI. Yet existing systems still tend to emphasize either physical grounding and action generation or open-ended agent reasoning and skill use. ME-Brain treats the two capability families as mutually reinforcing objectives and trains them within a unified Cognitive Core.

2.3 Embodied Operating Systems

Embodied capabilities and computational infrastructure. Embodied AI for scientific discovery relies on both model capabilities and supporting infrastructure. Prior work highlights edge computing, modular hardware, localized data pipelines, and open standards for practical deployment, particularly in resource-constrained environments [20, 21]. Unified embodied models and world–action modeling connect perception, decision-making, and execution through shared multimodal interfaces, integrated spatial reasoning and control, and representations of action-relevant physical states and outcomes [7, 19, 22]. At the computational level, unified inference infrastructure reduces implementation fragmentation across evaluation, cloud rollouts, edge serving, and on-robot deployment. Model adapters and shared execution mechanisms preserve model-specific logic while supporting low-latency interaction and scalable rollout generation [23]. These efforts establish model- and deployment-level foundations, whereas system-level research further addresses heterogeneous capability orchestration, interaction state maintenance, and execution monitoring.

Embodied system orchestration and continual adaptation. Hierarchical embodied frameworks connect high-level planning, skill invocation, and physical execution, exploring cross-embodiment collaboration, shared state, execution protocols, and outcome verification for multi-stage task coordination and supervision [5, 6]. Complementary embodied agent harnesses structure the perception–decision–execution loop through action and observation abstractions, tool interfaces, state maintenance, and structured feedback, enabling state tracking, feedback-driven replanning, and failure recovery in long-horizon tasks [24, 25]. For capability adaptation, reflection-based methods distinguish deficiencies in skill knowledge from execution errors to selectively refine procedural knowledge, while demonstration-based methods align human demonstrations with robot trajectories to rapidly acquire manipulation behaviors, both without task-specific parameter updates [8, 9]. Together, these efforts address task orchestration, closed-loop interaction, and skill refinement, providing complementary mechanisms for coordinating capability invocation, execution verification, and experience reuse within an Embodied OS.

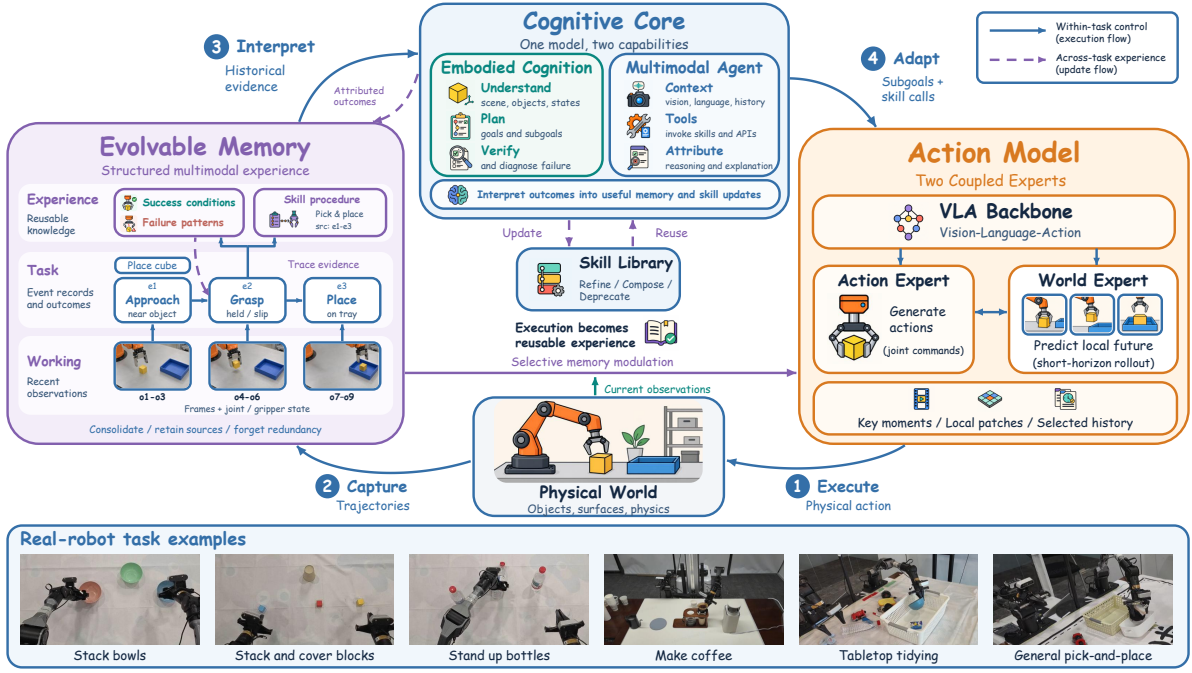


Figure 1. Experience-driven self-evolution framework of ME-Brain. Cognitive Core integrates embodied cognition with multimodal agent capabilities to plan subgoals, invoke skills, and interpret execution outcomes using current observations and historical evidence. Evolvable Memory consolidates observations and states into task events and reusable experience. Action Model executes the resulting subgoals through the coordination of the Action Expert and WM Expert. Together, the three modules connect within-task execution with cross-task experience evolution.

3 MachEmbodied-Brain Framework

Post-deployment self-evolution in embodied systems requires addressing three interdependent questions: how experience is acquired, how experience evolves, and how experience drives action execution. ME-Brain assigns these functions to three core modules—Evolvable Memory, Cognitive Core, and Action Model—and organizes them into a self-driven loop of action execution, experience acquisition, experience evolution, and improved action execution. This design allows the robot to continue learning from physical interaction without model retraining.

ME-Brain is designed to move beyond the train-and-freeze paradigm. After initial training, each task execution becomes an incremental source of system capability. The framework divides the system into memory, cognition, and action functions:

- **Evolvable Memory** structurally encodes and hierarchically consolidates multimodal execution streams into an external memory that is independent of the model context window, addressing how experience is acquired.
- **Action Model** translates language instructions into executable robot actions and serves as the low-level control component through which accumulated experience influences execution.
- **Cognitive Core** acts as the system hub, integrating embodied cognition with multimodal agent capabilities to interpret, consolidate, attribute, and transform experience into reusable skills.

The modules form a causal chain within the self-evolution loop. Action Model is the source of new experience: it executes subgoals issued by Cognitive Core, translates language intent into physical

action sequences, and continuously produces multimodal trajectories containing perceptual states, action decisions, and execution outcomes. Evolvable Memory is the consolidation layer: it encodes and archives these trajectories in an external memory graph, converting a one-time execution into persistent knowledge that can be retrieved and transferred. Cognitive Core is the experience refinement layer: it retrieves historical evidence, attributes success conditions and failure modes, and consolidates repeatedly validated strategies into new skills that can be invoked by Action Model.

The output of each module becomes the input to the next, forming a complete causal chain in which execution produces experience, experience is recorded, records are transformed into capability, and evolved capability improves future execution. The autonomous operation of this chain allows ME-Brain to expand its capability boundary through continued deployment rather than external retraining.

4 Evolvable Memory

Reactive action models such as OpenVLA generate the next action from the current image observation and language instruction, without explicitly conditioning on observation history [2]. In long-horizon tasks, however, the evolving states of objects, accumulated outcomes of past actions, causes of failure, and dynamic environmental changes often exhibit strong causal dependencies. Under partial observability, different underlying states can produce indistinguishable current observations, leading to state aliasing and limiting policies that treat the current observation as a complete state [26]. Furthermore, embodied systems operating over extended periods must maintain environmental states and task experiences across time from continuous observations, requiring long-term multimodal memory to guide action generation. Maintaining such memory introduces information redundancy and computational overhead, which constrain the temporal coverage, information density, and retrieval efficiency of existing memory approaches. Existing memory and retrieval systems use textual or structured semantic abstractions: SimpleMem consolidates interactions into compact semantic memory units [27], MemoryBank maintains conversation records and hierarchical event summaries [28], and GraphRAG organizes document information into entity graphs and community summaries [29]. MIRIX extends memory to multimodal inputs through specialized memory components [30]. These designs motivate a complementary requirement for embodied action generation: retaining fine-grained visual, spatial, and action-state information alongside semantic abstractions.

To address these limitations, we propose *Evolvable Memory*, an evolvable multimodal memory module for embodied intelligence that combines global causal modeling with local Markovian action modeling. By incorporating key historical information, including spatial locations, inter-object relationships, and execution states at the time of an event, Evolvable Memory provides historical conditions that are difficult to recover from the current observation alone, thereby improving the understanding and execution of long-horizon tasks. As illustrated in Figure 2, the memory system progressively organizes multimodal interaction history into hierarchical representations and selectively retrieves information according to the requirements of downstream modules.

To support fine-grained memory over extended interaction horizons, Evolvable Memory incorporates two key mechanisms:

- **Multimodal Embodied Memory Storage.** In addition to high-level semantic memories, the module preserves low-level visual and action memories to provide fine-grained guidance for task planning and execution. Hierarchical storage and evolutionary compression consolidate memories into compact, generalizable knowledge, allowing them to persist beyond a single context window and to be continuously updated throughout task execution, ultimately forming a memory knowledge base with

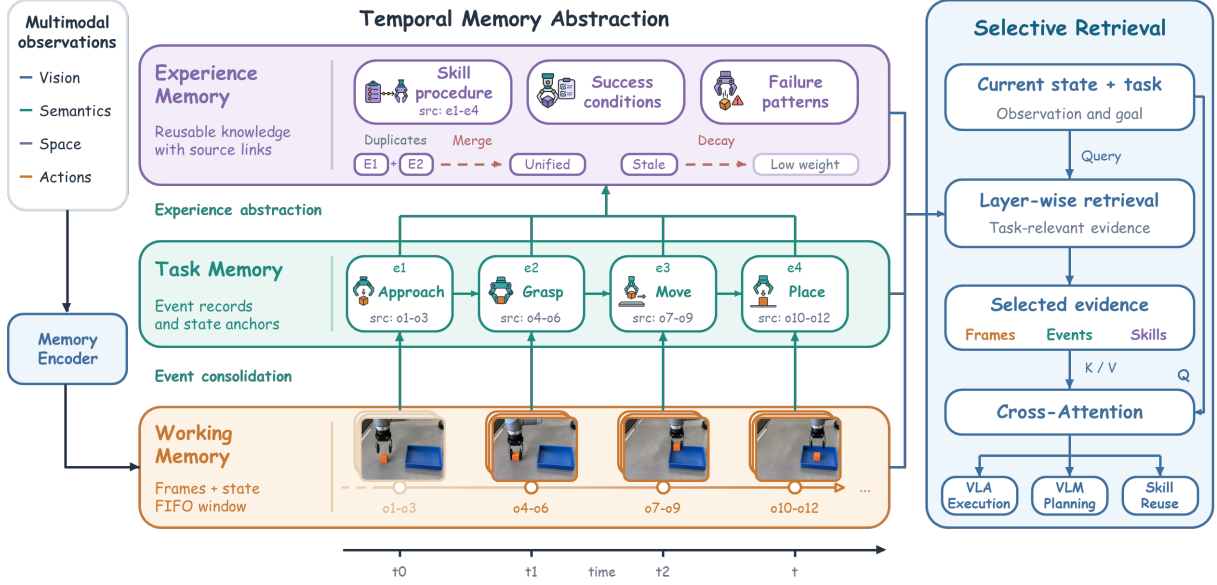


Figure 2. Multimodal hierarchical memory and selective access. Multimodal observations are encoded by a memory encoder and written to working memory, consolidated into task memory through event aggregation, and further abstracted into skill procedures, success conditions, and failure patterns. Provenance links preserve associations with supporting evidence across memory levels, while first-in, first-out (FIFO) eviction, duplicate merging, and downweighting of stale information support memory maintenance. The current state and task guide hierarchical retrieval, and the selected evidence is incorporated through cross-attention to support action execution, task planning, and experience reuse.

no fixed theoretical upper bound. Meanwhile, more efficient memory retrieval enables more efficient memory injection, reducing its computational complexity from $O(n^2)$ to $O(n)$ relative to conventional memory injection approaches.

- **Continual Memory Evolution.** To support long-term embodied operation, we design a memory evolution mechanism that enables continual accumulation and dynamic updates. Through hierarchical storage, evolutionary compression, and on-demand retrieval and injection, embodied memories are continuously recorded and evolve from the bottom up, progressively consolidating raw observations into events and skill-related experience. As memory evolves, the Action Model and Cognitive Core develop an increasingly detailed understanding of the environment and their own capability boundaries. The former uses its understanding of these boundaries to reduce trial and error, while the latter draws on historical failure cases to identify root causes and resume execution more efficiently. Continual memory evolution enables the overall system to become increasingly proficient over extended operation.

4.1 Memory Storage and Evolution

4.1.1 Multimodal Embodied Memory Storage

Memory systems for embodied tasks must support both high-level task planning and fine-grained action generation. High-level semantics can summarize task objectives, execution stages, and successful or failed experiences, whereas concrete manipulation additionally depends on object locations, inter-object relationships, state changes before and after actions, and the temporal order in which these changes occur. Retaining only textual summaries of events can therefore result in the loss of execution-relevant information. Accordingly, our memory system preserves visual observations, spatial states, and action information alongside high-level semantic memories, enabling historical experience to guide task understanding and concrete execution at different levels of granularity.

To construct these memory representations, we introduce an *Embodied Model* specifically adapted to memory-related tasks as a specialized memory expert. The model receives multimodal observations and action states collected during task execution, extracts semantic information from key events, updates embodied state representations, and generates structured memory nodes. Each node organizes retrievable textual semantics together with the associated visual context, object locations, inter-object relationships, and action outcomes, preserving the execution-level context underlying summarized experience. Through the links between nodes in the hierarchical memory structure, the system can use high-level semantics for planning while accessing fine-grained historical conditions relevant to the current operation.

Consider the task of picking up a cup, replacing the tablecloth, and then returning the cup to its original position. The final placement action depends on the cup’s location at the beginning of the task. A semantic summary such as “the cup was originally on the table” captures the general event but does not specify a sufficiently precise placement target. A multimodal memory node preserves both the original spatial location and the associated visual context, allowing the system to retrieve the relevant historical state when executing the instruction to “return the cup to its original position” and to provide placement conditions to the base action model through memory injection. This example illustrates the complementary roles of semantics and embodied states: the former supports the interpretation of task requirements, while the latter provides the concrete information needed to generate actions that satisfy those requirements.

4.1.2 Memory Evolution

Evolvable Memory supports memory retention over month-scale time horizons, allowing historical information to persist across individual tasks and beyond model context windows. During long-horizon tasks, object states, action outcomes, and environmental changes continuously accumulate, collectively forming the historical conditions required for subsequent decisions. Relying solely on current observations and a limited context buffer makes it difficult to maintain this information over extended periods. Evolvable Memory therefore combines hierarchical storage, evolutionary compression, and on-demand retrieval and injection to retain long-term history while accessing the information required for current decisions, extending the temporal scope of memory while controlling maintenance and inference overhead.

Across temporal scales, Evolvable Memory organizes information into short-term, mid-term, and long-term memory and progressively consolidates information generated during execution through evolutionary compression. Short-term memory retains high-frequency, temporally dense multimodal states, providing recent history for local action execution. Mid-term memory aggregates key events and relationships between execution stages, providing task-level context for planning. Long-term memory stores abstracted success conditions, failure patterns, and semantic experience, supporting retrieval and reuse across tasks. As execution progresses, short-term records are consolidated into mid-term events, and experiences with reuse potential are further abstracted into long-term memory. Mid-term and long-term nodes retain links to the underlying fine-grained information, enabling the system to form compact experience representations while preserving access to specific execution conditions.

In terms of memory organization, Evolvable Memory represents each complete task as a parent node, with subtasks and key events stored as associated child nodes, forming an external memory structure maintained independently of the model’s context window. Parent nodes summarize task objectives and overall execution, while child nodes record specific operations, local state changes, and success or failure outcomes. This structure allows the system to trace the relationships between an outcome and the relevant operations and environmental states along the task hierarchy. When historical conditions cannot be obtained directly from current observations, the system retrieves the relevant information through associated nodes, supporting cross-stage planning, action execution, and failure recovery.

To efficiently incorporate continuously accumulated memories into current decisions, Evolvable Memory selects the retrieval level according to the requirements of each downstream module and then locates relevant nodes within that level. The action model retrieves fine-grained states from short-term memory based on spatiotemporal proximity and the objects involved in the current operation. The task planning module retrieves key events and multimodal features from mid-term memory based on task and subtask relationships. The experience summarization module uses semantic similarity to identify reusable experience in long-term memory. This hierarchical retrieval mechanism selects decision-relevant information from the complete history, reducing indiscriminate scanning and the injection of irrelevant content.

4.2 Edge Deployment, Privacy, and Security

Long-term embodied memory continuously records environmental observations, user behavior, and task histories, and its deployment must therefore account for both local resource constraints and data privacy requirements. In home, in-vehicle, and industrial settings, on-device memory processing can reduce the transmission of sensitive information between devices and the cloud while shortening the access path for memory generation and retrieval. Evolvable Memory therefore adopts on-device operation as the deployment foundation for long-term memory, allowing its capabilities to directly support local task execution. Built on *Cognitive Core*, Evolvable Memory has been ported to M100 and optimized for this platform, supporting on-device memory generation, storage, evolution, and retrieval. This deployment enables memory-related processing to be performed locally, providing a foundation for maintaining sensitive data on the device. Combined with hierarchical storage and on-demand access, the system organizes and uses memories across different temporal scales on edge devices, supporting long-term embodied applications in real-world environments.

5 Cognitive Core

Cognitive Core provides the cognitive capabilities of ME-Brain and coordinates with Evolvable Memory and Action Model to form a complete embodied system [31]. We develop a balanced training approach that jointly trains embodied-cognition and multimodal-agent experts through SFT and RL [32], followed by MOPO distillation to balance the two capability families. This design allows the model to develop physical understanding and intelligent decision making within a unified model.

As the central hub of ME-Brain, Cognitive Core retrieves historical evidence from Evolvable Memory to improve current decisions and writes causally attributed execution outcomes back into memory to drive continued experience accumulation. It also sends subgoals and tool or skill invocation instructions to Action Model and injects recovery strategies when execution deviates from the plan. The three modules therefore form a closed loop of cognition, memory, and action that extends self-evolution from individual executions to experience shared across tasks.

5.1 Unified Capabilities

5.1.1 Embodied Cognitive Capability

Embodied cognition is organized around the physical execution loop. Embodied scene understanding constructs an actionable state representation from perception; embodied task planning decomposes high-level objectives into executable subtasks; embodied action understanding relates actions to changes in the physical state; and verification and replanning diagnose failures and revise the plan. Together, these

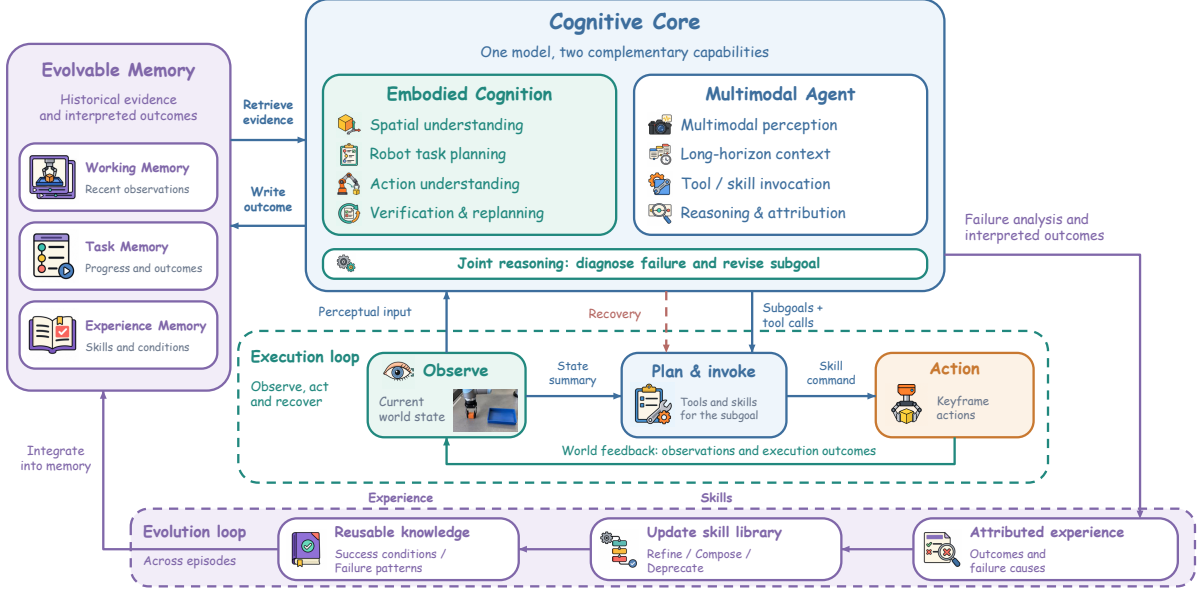


Figure 3. Execution and evolution loop of Cognitive Core. The unified cognitive model combines embodied cognition with multimodal agent capabilities to interpret the current state using historical evidence, plan subgoals, and invoke skills, while Action model executes the resulting actions. Observation feedback supports verification, failure diagnosis, and replanning. Causally attributed outcomes support skill updates and experience abstraction and are written back to memory for future reuse.

capabilities form an observe-plan-act-replan loop for continuous and robust long-horizon execution.

1. **Embodied scene understanding.** A conventional VLM can describe visible content, whereas Cognitive Core must additionally determine what is actionable, where to grasp, and where an object should be placed. It connects visual semantics, spatial geometry, and robot affordances to form an action-oriented state representation. Given an observation o_t and instruction l , the model produces an affordance-aware task decomposition

$$\mathcal{A}(o_t, l) = \{(r_i, \text{aff}_i, \text{pos}_i)\}_{i=1}^N, \quad (1)$$

where r_i describes an actionable region, aff_i denotes its affordance type, and pos_i is a 2D image coordinate.

2. **Embodied task planning.** The planner decomposes a high-level instruction into executable and verifiable subtasks while jointly considering the objective, environment state, available skills, and execution feedback. Given instruction l , observation o_t , robot state s_t , and memory M , it generates or updates a sequence of subtasks:

$$\{g_1, g_2, \dots, g_n\} = f_{\text{plan}}(l, o_t, s_t, M). \quad (2)$$

Each g_i specifies an operation intent, target object, execution constraints, and observable completion conditions for Action Model.

3. **Embodied action understanding.** Beyond predicting actions, the model captures the causal relation between an action and its effect on the environment. Given state s_t and proposed action a_t , it predicts a semantic description of the successor state,

$$p(s_{t+1} \mid s_t, a_t), \quad (3)$$

including object displacement, contact changes, and task-stage progression. For example, completing a grasp requires observing gripper closure, object lift, and the absence of slippage, rather than merely issuing a grasp command. Cognitive Core uses this interpretation to select the next subtask or trigger recovery.

4. **Verification and replanning.** When the current observation o_t violates an expected postcondition, the model first identifies the failure cause and then updates the task-progress graph and produces a corrected plan. The diagnosis is also written to memory as context for future reasoning. Diagnosis quality directly determines the quality of the failure experience stored in Evolvable Memory.

5.1.2 Multimodal Agentic Capability

Multimodal agent capabilities turn experience into reusable behavior. Multimodal understanding determines the semantic quality of each experience record; long-horizon task understanding maintains coherent state across steps and tasks; tool and skill invocation consolidates experience into reusable composite skills; and reasoning and attribution provide the basis for extracting useful experience from execution outcomes. Together, they form a progression from experience acquisition to capability evolution.

1. **Multimodal understanding.** The model extracts manipulation semantics from observations, interprets task intent from language, and aligns visual evidence with language reasoning. This capability determines the information density of memory nodes and the reliability of subsequent retrieval and cross-task analogy.
2. **Long-horizon task understanding.** The system maintains coherent task state across multiple subgoals and across tasks instead of making isolated single-step decisions. A longer retrievable history supports broader experience consolidation and the transfer from solving one task instance to solving a class of related tasks.
3. **Tool and skill invocation.** Atomic embodied capabilities are exposed through semantic interfaces that can be invoked on demand. Rather than directly producing low-level action sequences, the agentic loop orchestrates embodied tools and skills, while an Embodied Skill Library stores validated composite skills for cross-task reuse. A continual evolution protocol refines preconditions, composes repeatedly successful atomic sequences into new skills, and replaces skills that fail persistently.
4. **Reasoning and attribution.** After an execution failure, the system determines whether the cause lies in perception, planning, or action precision and generates a targeted correction. Stronger attribution improves both the labels attached to failure examples in Evolvable Memory and the decisions made by the skill-update protocol, thereby setting the upper bound on experience evolution.

5.2 Collaboration with Other Modules

Cognitive Core and Evolvable Memory form a bidirectional read–write loop. During reading, Cognitive Core retrieves key nodes from mid-term event memory using subtask relationships and matches reusable success conditions and failure patterns from long-term experience memory using semantic similarity. During writing, it interprets each execution outcome and produces causally annotated records describing the failure cause, state-transition process, and subtask completion status, which are written to the appropriate memory level. Cognitive Core is therefore both a consumer of memory and a primary producer of high-quality memory.

Collaboration between Cognitive Core and Action Model operates at two levels. For execution, the *Observe* node sends the current world state to Cognitive Core; the embodied-cognition component performs spatial

and state analysis, and the multimodal-agent component converts the result into subgoals and tool or skill calls for the *Plan & Invoke* node. Action Model then executes the corresponding keyframe action sequence. For closed-loop recovery, action outcomes and new observations continuously return to the Observe node. Verification and replanning compare this feedback with expected outcomes, diagnose deviations, and inject a targeted recovery strategy into *Plan & Invoke* without restarting the complete task. This loop preserves execution continuity and robustness in unexpected states.

6 Action Model

Many vision-language-action (VLA) models generate an action sequence from the current visual observations $\mathbf{O}_t$, robot state $\mathbf{s}_t$, and language instruction ℓ , following a policy of the form $\pi_\theta(\mathbf{O}_t, \mathbf{s}_t, \ell)$ [3, 4]. World-action models extend this formulation by jointly predicting visual futures and actions [16], while memory-augmented policies incorporate temporal context [12, 14]. These extensions can introduce substantial redundancy when computation is allocated across densely sampled action sequences, full future images, or unfiltered observation histories. Yet task-relevant information in robotic manipulation is unevenly distributed: many consecutive frames contain little motion, large image regions are unrelated to the immediate interaction, and only a small subset of historical states determines the next action. Processing these inputs without distinguishing their relevance consumes computation and can dilute the signals associated with contact, grasping, and placement.

Motivated by this observation, we introduce **Focus-VLWA**, an interaction-event-centric *Vision-Language-World-Action* model. Its central principle is to allocate modeling capacity to the key moments, interaction regions, and historical evidence associated with the next physical interaction, rather than treating all temporal and spatial positions alike. As shown in Figure 4, Focus-VLWA extends a VLA with a world-model expert and an action-conditioned memory pathway.

Historical evidence and future-world representations are incorporated without simply lengthening the model’s input and output sequences. Temporally, dense control trajectories are represented by keyframes corresponding to salient physical changes. Spatially, future prediction is restricted to neighborhoods around the next interaction event. For memory, action-conditioned retrieval selects relevant historical evidence without expanding the vision-language backbone’s context. Together, these mechanisms connect the historical evidence needed for the current decision, the location and temporal offset of the next interaction, and the motion required to approach its target state.

6.1 Event-Centric World-Action Modeling

Focus-VLWA adopts the vision-language backbone of $\pi_{0.5}$ [4] to encode the current task state. The model receives current images from the head, left-wrist, and right-wrist cameras, the task instruction, and discretized robot state. PaliGemma [1] produces contextual representations of scene semantics, manipulation targets, and proprioceptive state. Historical memory provides visual evidence from the past, while the world-model expert predicts future changes associated with the next interaction event.

We represent the next interaction event as

$$e_t = \left(k_t^e, c_t^H, c_t^L, c_t^R, \Delta_t, \mathbf{a}_t^e \right), \quad (4)$$

where k_t^e is the keyframe of the next event, such as a grasp, contact, or placement; c_t^H , c_t^L , and c_t^R identify its spatial cells, termed *EventCells*, in the three camera views; Δ_t is the number of keyframes remaining

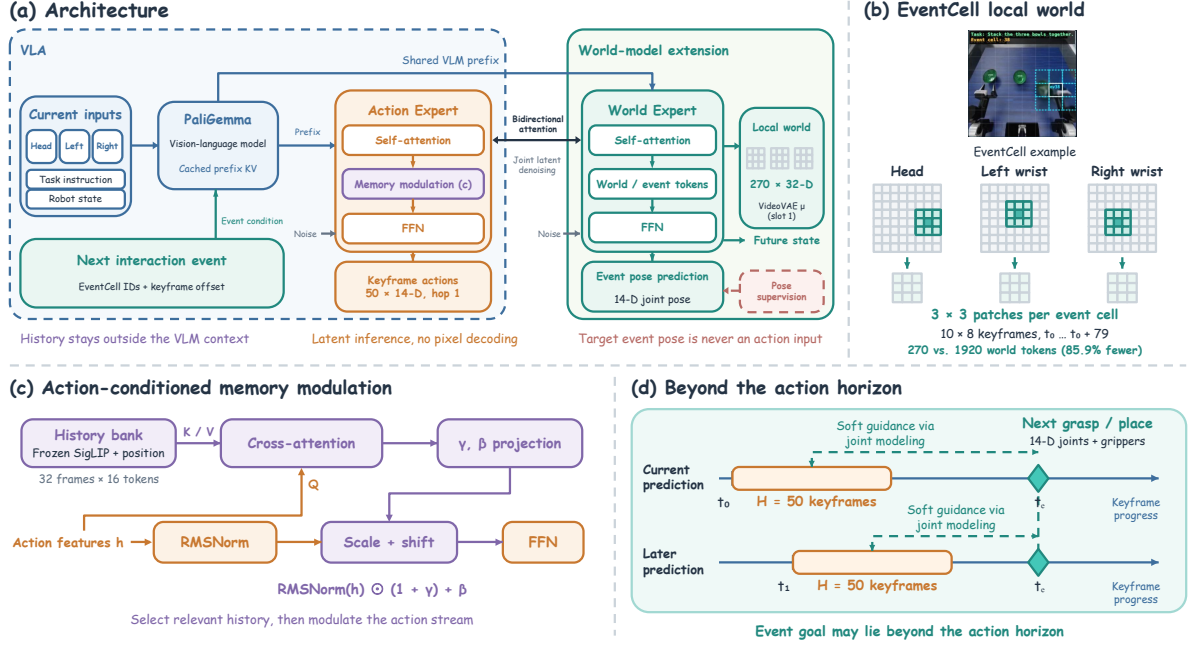


Figure 4. Focus-VLWA architecture and key mechanisms. (a) A VLA and a world-model expert share a vision-language prefix and exchange information through bidirectional attention for joint latent-space denoising. (b) EventCell restricts future-world prediction to multi-view interaction neighborhoods. (c) Action features query an external history bank and modulate normalized action representations without extending the backbone context. (d) Fixed-length action windows advance with replanning, while the next event target may lie beyond the current horizon and provide guidance through joint modeling. Ground-truth event poses are supervision targets, not action inputs.

until the event; and $\mathbf{a}_t^e \in \mathbb{R}^{14}$ is the target joint-and-gripper configuration at that event.

The event serves as an alignment anchor for world prediction and action generation. EventCell specifies where future-world modeling should focus, the event offset situates the current state on the event timeline, and the event pose provides the physical target state. World and action predictions are thus aligned around a shared interaction instead of an undifferentiated visual future. The event target may lie beyond the fixed action horizon (Figure 4(d)), providing goal guidance through joint modeling without requiring the current action chunk to reach the event immediately.

In the current implementation, EventCell IDs are included in the action model’s prompt as structured conditions. The ground-truth event pose $\mathbf{a}_t^e$ is used only for supervision and is not supplied to the action expert. We denote the event-conditioning information available to the model by $\tilde{e}_t$, to distinguish it from the complete event annotation e_t in Equation (4). In particular, $\tilde{e}_t$ excludes $\mathbf{a}_t^e$. The action expert must infer the motion toward the event from the current observations, task semantics, historical evidence, and available event conditions, rather than reading the target pose directly. An action event here is a low-level physical event defined by robot motion and interaction-state changes, not a high-level semantic event with a human-written description. Event annotations and conditioning should not themselves be interpreted as an online event-detection capability.

Along the temporal dimension, Focus-VLWA predicts event-driven keyframe actions instead of densely sampled control targets at a fixed rate. Adjacent targets in a dense trajectory often differ only slightly, so a fixed-length prediction window can be dominated by stationary states or small-amplitude motion and cover limited task progress. Waypoint-based imitation learning has previously shown the value of compact trajectory representations for reducing the learning horizon [33]. Focus-VLWA uses joint and gripper changes to organize the action sequence on a keyframe timeline shared with future-world supervision.

Specifically, a new action keyframe is retained when a robot joint changes by more than 1° or a gripper displacement exceeds 5 mm. This resampling compresses the dense control trajectory into a sequence of salient physical changes. Given the current keyframe index t_0 , the model predicts

$$\hat{\mathbf{J}}_{t_0} = [\hat{\mathbf{j}}_{t_0}, \hat{\mathbf{j}}_{t_0+1}, \dots, \hat{\mathbf{j}}_{t_0+49}], \quad \hat{\mathbf{j}}_i \in \mathbb{R}^{14}. \quad (5)$$

These 50 positions represent consecutive keyframe targets, not 50 motor commands at a fixed temporal sampling interval. Each training sample is anchored at a current keyframe, with the corresponding action supervision

$$\mathbf{J}_{t_0} = [\mathbf{j}_{t_0}, \dots, \mathbf{j}_{t_0+49}] = \mathbf{j}_{14}[t_0 : t_0 + 50], \quad (6)$$

where the slice uses an exclusive upper bound. Removing stationary and repeated states concentrates supervision on meaningful changes in robot configuration and interaction state. For a fixed output length, the resulting representation can cover more task progress and a longer elapsed-time span, linking the motion stages leading toward the next grasp or placement.

Future-world supervision uses the same keyframe timeline. We divide keyframes t_0 through $t_0 + 79$ into ten world chunks, each containing eight consecutive keyframes:

$$[t_0 + 8i, t_0 + 8i + 7], \quad i = 0, \dots, 9. \quad (7)$$

Instead of making a dense, stride-one future-video prediction at every keyframe, the world branch uses ten temporally compressed representations to cover 80 keyframes. The action branch predicts consecutive keyframe configurations at hop one, while the world branch describes environmental changes at a lower temporal density. Both are aligned to physical task progress rather than a frame-by-frame replay of the original control stream.

Along the spatial dimension, Focus-VLWA introduces **EventCell local-world prediction**. Video-based world-action models such as DreamZero jointly model future visual observations and actions [16]. Recent work also explores action-centered modeling to reduce the inference overhead of video prediction [17]. Our design focuses specifically on the spatial support of world supervision: for manipulation, grasp points, contact regions, gripper states, and target placement locations are often more informative than background regions far from the interaction.

We partition the visual latent grid of each camera view into 8×8 cells and locate the EventCell associated with the next interaction. Rather than predicting the full grid, the world-model expert predicts only the 3×3 neighborhood centered on that cell. Each future world chunk therefore contains 27 local visual tokens across the head and two wrist views, with nine tokens per view. The head view provides scene context around the event, while the wrist views capture local details near the grippers and contact regions. Each patch is represented by a 32-dimensional vector from temporal slot 1 of the VideoVAE posterior mean μ . Across ten world chunks, the local-world target contains 270 such tokens.

Under the same single-temporal-slot representation, predicting the complete 8×8 grid for all three views and ten chunks would require

$$N_{\text{full}} = 10 \times 3 \times 64 = 1920, \quad N_{\text{local}} = 10 \times 3 \times 9 = 270. \quad (8)$$

EventCell thus reduces the number of world tokens by approximately 85.9%. This is a reduction in representation length, not a measured end-to-end latency reduction. Unlike uniform downsampling of the full future scene, the spatial selection is tied to the next interaction event, preserving localized changes relevant to the action while limiting the world-modeling budget.

The action and world-model experts jointly model keyframe actions, local-world tokens, future proprioceptive states, and event targets. The current camera views, task instruction, and robot state form a shared vision-language prefix; the experts exchange information through bidirectional attention during latent denoising. With historical memory $\mathbf{M}_{t_0}$ and event conditions $\tilde{e}_{t_0}$, this joint predictor is written as

$$F_{\theta}(\mathbf{O}_{t_0}, \mathbf{s}_{t_0}, \ell, \mathbf{M}_{t_0}, \tilde{e}_{t_0}). \quad (9)$$

Its training objective combines

$$\mathcal{L} = \lambda_a \mathcal{L}_{\text{action}} + \lambda_w \mathcal{L}_{\text{local-world}} + \lambda_s \mathcal{L}_{\text{state}} + \lambda_e \mathcal{L}_{\text{event}}, \quad (10)$$

where $\mathcal{L}_{\text{action}}$ supervises future keyframe actions, $\mathcal{L}_{\text{local-world}}$ supervises the VideoVAE latents around EventCells, $\mathcal{L}_{\text{state}}$ supervises future proprioceptive states, and $\mathcal{L}_{\text{event}}$ promotes consistency among event locations, temporal offsets, and target configurations.

This formulation makes local dynamics prediction part of action learning. In addition to matching demonstration trajectories, action representations are trained jointly with changes in object motion, contact relationships, and gripper states near the next interaction. The world branch therefore aims to predict action-relevant local consequences, rather than reconstruct a visually complete future video as an independent end task.

6.2 Action-Conditioned Memory Modulation

EventCell focuses prediction on future interaction regions, but partially observable tasks also require evidence that is no longer visible. A target may be occluded, an operation may need to be repeated a specified number of times, or an object may need to be returned to its initial position. Identical current observations can therefore require different actions depending on the preceding trajectory. Memory-augmented VLA methods address related temporal dependencies through explicit memory or multi-scale historical representations [12, 14].

A straightforward way to expose history to a policy is to concatenate historical visual tokens with the current vision-language context. Under dense attention, increasing this token history increases the backbone’s attention workload and context storage; repeatedly processing the enlarged context can also increase inference latency. Focus-VLWA instead maintains an independent visual memory bank and lets the action expert query it directly, leaving PaliGemma’s main context unchanged.

For each action prediction, we uniformly sample up to 32 head-camera frames from the beginning of the trajectory to the current time. A frozen SigLIP encoder [34], followed by spatial pooling, produces 16 visual tokens per frame, giving at most 512 historical tokens per read. Each visual feature is concatenated with temporal and two-dimensional spatial position embeddings and projected to a 1024-dimensional memory representation:

$$\mathbf{m}_{i,j} = f_{\text{mem}}\left(\left[\mathbf{v}_{i,j}; \mathbf{e}_i^{\text{time}}; \mathbf{e}_j^{\text{space}}\right]\right), \quad (11)$$

where i indexes historical frames and j indexes pooled spatial positions. These tokens form $\mathbf{M}_t$ but are not appended to the current image and language prefix.

At layer l of the action expert, each future action position provides a query, while the historical memory provides keys and values:

$$\mathbf{c}^l = \text{Attn}\left(Q_l \mathbf{h}^l, K_l \mathbf{M}_t, V_l \mathbf{M}_t\right). \quad (12)$$

The attention weights select evidence relevant to the evolving action representation. Different action

positions can attend to different historical states, such as an object’s initial location, the most recent contact, or evidence of previously completed repetitions. The retrieved context is projected to feature-wise scale and shift parameters,

$$\left(\gamma^l, \beta^l\right) = g_l(\mathbf{c}^l), \quad (13)$$

which modulate the normalized action features after self-attention and before the feed-forward network:

$$\tilde{\mathbf{h}}^l = \text{RMSNorm}(\mathbf{h}^l) \odot \left(1 + \gamma^l\right) + \beta^l. \quad (14)$$

This operation uses RMSNorm [35] and a feature-wise affine conditioning form related to FiLM [36]. Our design derives the modulation parameters from action-conditioned retrieval over a separate visual history bank and applies them within the action expert, rather than adding historical tokens to the backbone context.

Cross-attention determines which evidence to retrieve, while the scale and shift parameters control how that evidence changes the action features. Memory thus influences action generation at different network depths and future action positions, rather than serving only as a static input prompt.

Let N_c , N_m , and N_a denote the numbers of current-context, memory, and action tokens. For fixed feature dimensions and network depth, concatenating memory into a densely attended backbone context yields an attention-score complexity of

$$O\left((N_c + N_m)^2\right). \quad (15)$$

Keeping the backbone context unchanged and reading memory through the action expert instead gives

$$O(N_c^2) + O(N_a N_m), \quad (16)$$

omitting costs that do not vary with the memory length, such as action self-attention at a fixed horizon. Thus, for fixed N_a , the additional memory attention scales linearly with the number of retrieved tokens. This comparison concerns attention computation, not a claim that total system latency follows the same asymptotic ratio. In the current implementation, the 32-frame sampling cap bounds the token budget while allowing the sampled history to span a longer trajectory.

During training, historical memory supplies past evidence and local-world supervision constrains future consequences. Future-world latents, robot states, and actions are trained with a joint flow-matching objective [37]. Asynchronous noise times expose the model to modalities at different stages of denoising. The vision-language backbone and VideoVAE remain frozen; training updates the action and world-model experts, memory connectors, and newly introduced event and action prediction modules. This preserves the pretrained vision-language and video priors.

At deployment, the vision-language prefix and historical memory representations are computed once per observation update and reused across action-denoising steps. The action-serving path operates in latent space without decoding future video pixels. EventCell latents still participate in joint world-action inference, but the system avoids pixel-level reconstruction of the full scene and prediction of spatial regions outside the selected neighborhoods. Skipping pixel decoding therefore does not mean removing the world-model expert from action inference.

In summary, Focus-VLWA organizes action generation around selective computation for physical interactions. Keyframe actions reduce temporal redundancy, EventCell concentrates future-world modeling on local interaction regions, and action-conditioned modulation retrieves relevant history without continually expanding the vision-language context. Historical evidence, local future changes, and motion toward the

Table 1. Results on the embodied benchmark. Average values follow the corresponding evaluation runs. Dashes denote unavailable results. The best result on each benchmark is boldfaced, including ties.

Subcategory	Benchmark	ME-VLM 4B	ME-VLM 35B-A3B	Hy-Embodied 30B-A3B	RynnBrain1.1 9B	Mimo-Embodied 7B	PhysBrain 1.5 8B
	Average	63.4	70.9	60.2	62.1	54.3	62.7
Physical Understanding	CV-Bench	89.5	91.7	89.7	88.2	88.8	90.0
	VABench-Point	67.0	75.4	59.7	20.3	33.3	65.2
	VABench-Visual	81.6	82.6	79.7	87.6	66.9	89.8
	RefCOCO-testA	85.9	87.8	49.4	89.5	74.5	60.9
	RefCOCO-testB	76.6	80.3	56.2	81.6	67.4	62.8
	ERQA	45.8	54.3	60.8	47.5	46.8	52.8
	MindCube	–	91.1	70.0	86.9	35.2	86.2
	SPARBench	57.3	67.6	53.4	51.1	41.2	53.8
	SPBench-MV	69.9	83.5	59.8	74.4	49.4	78.8
	SPBench-SI	70.6	71.7	50.5	59.0	44.8	77.7
	MMSI-Bench	33.3	46.4	41.8	47.0	29.6	41.0
	ViewSpatial-Bench	57.8	62.7	53.3	54.2	40.5	62.5
	EmbSpatial-Bench	81.0	83.1	82.7	81.9	76.2	81.8
	RefSpatial-Bench	46.9	63.4	53.4	67.2	48.0	50.9
	RoboSpatial-Home	67.7	74.0	69.4	69.1	61.8	73.9
Task Planning	Cosmos	65.7	76.9	66.9	56.1	56.8	72.8
	EgoPlan2	55.3	64.5	49.6	43.5	43.0	62.1
	RoboBench-Planning	40.1	53.7	54.9	59.8	57.2	42.2
Action Execution	VSIBench	63.2	71.2	58.9	74.9	48.5	61.9
	SITE-Bench-Video	65.6	72.9	69.2	68.9	59.0	67.5
	RoboBench-Perception	41.8	52.2	55.9	40.5	34.9	31.8
	RoboBench-Affordance	47.0	60.7	61.7	33.0	36.7	25.6
Correction	RoboFail-Execution	79.7	86.3	62.8	77.1	72.6	77.8
	RoboFail-Planning	56.7	63.3	53.3	63.3	63.3	56.7
	RoboFAC	74.7	75.7	51.0	55.4	61.2	64.8
	RoboBench-Error	–	52.7	53.0	42.2	33.5	34.1

next event are combined in a single vision-language-world-action framework, with computation directed toward the information that matters for control.

7 Experiments

7.1 Evaluation of Cognitive Core

7.1.1 Embodied Benchmark

The embodied evaluation spans 26 benchmarks covering physical and spatial understanding, task planning, action execution, and execution correction. We compare ME-VLM at 35B-A3B and 4B scales with Hy-Embodied 30B-A3B, RynnBrain1.1 9B, Mimo-Embodied 7B, and PhysBrain 1.5 8B.

ME-VLM 35B-A3B achieves the highest overall average of 70.9, exceeding the strongest comparison model by 8.2 points and obtaining the best or tied-best result on 14 of the 26 benchmarks. Its advantages are concentrated in the capabilities that connect physical understanding to executable behavior. The model reaches 91.1 on MindCube, 83.5 on SPBench-MV, 83.1 on EmbSpatial-Bench, and 74.0 on RoboSpatial-Home, establishing strong spatial reasoning across complementary task formats. It also leads on Cosmos and EgoPlan2 with 76.9 and 64.5, and obtains 72.9 on SITE-Bench-Video.

The execution-correction results further validate the closed-loop cognitive design. ME-VLM achieves 86.3

Table 2. Results on the agent benchmark. The average follows the benchmark suite shown in the table. Dashes denote unavailable results. The best result on each benchmark is boldfaced, including ties.

Subcategory	Benchmark	ME-VLM 4B	ME-VLM 35B-A3B	Hy-Embodied 30B-A3B	RynnBrain1.1 9B	Mimo-Embodied 7B	PhysBrain 1.5 8B
	Average	63.1	72.5	61.2	62.9	54.1	46.6
Multimodal Understanding	BLINK	58.9	67.7	66.2	53.8	56.3	64.7
	MMStar	72.5	77.3	76.1	68.7	68.7	62.5
	MVBench	64.1	71.9	67.6	62.3	56.7	60.8
	VideoMME	55.6	60.3	58.2	55.4	25.1	52.2
	RealWorldQA	79.0	79.7	76.6	39.5	67.1	71.5
Tool or Skill Invocation	BFCL-V4	40.5	63.2	–	63.2	51.5	–
	TAU2-Bench	50.1	63.9	–	75.6	15.0	–
Long-Horizon Planning	ClawEval avg ³	–	58.0	–	62.4	37.0	–
	ClawEval pass ³	–	31.8	–	37.2	6.0	–
Reasoning	GPQA	67.2	78.7	46.5	75.3	54.9	35.0
	AIME25	47.5	72.9	26.7	70.4	48.3	3.8
	MMLU-Pro	75.4	82.8	74.3	81.3	71.6	46.7
	MMMU-Pro	58.4	69.3	64.8	60.8	52.1	35.1
	MathVision	59.6	74.3	66.1	61.4	54.5	19.6
Instruction Following	IFEval	81.9	86.9	82.1	83.6	65.6	79.1
	IFBench	37.4	48.3	29.3	42.5	27.9	27.6

on RoboFail-Execution, 63.3 on RoboFail-Planning, and 75.7 on RoboFAC. Together with the broad gains in physical and spatial understanding, these results demonstrate that Cognitive Core connects perception, planning, execution understanding, and recovery in a single model rather than optimizing an isolated embodied skill.

7.1.2 Agent Benchmark

The agent evaluation covers multimodal understanding, tool and skill invocation, long-horizon planning, reasoning, and instruction following. We compare ME-VLM at 35B-A3B and 4B scales with Hy-Embodied 30B-A3B, RynnBrain1.1 9B, Mimo-Embodied 7B, and PhysBrain 1.5 8B.

ME-VLM 35B-A3B achieves the highest overall average of 72.5, outperforming the strongest comparison model by 9.6 points. The 4B model also reaches 63.1, surpassing every comparison model in overall average. The 35B-A3B model ranks first on all five multimodal understanding benchmarks and all five reasoning benchmarks, demonstrating that embodied specialization preserves a broad multimodal and reasoning foundation. The model reaches 71.9 on MVBench, 60.3 on VideoMME, 78.7 on GPQA, 72.9 on AIME25, and 74.3 on MathVision.

The same unified model also delivers strong structured interaction and instruction execution. It matches the best result on BFCL-V4 at 63.2 and achieves 86.9 on IFEval and 48.3 on IFBench, leading the strongest comparison model by 3.3 and 5.8 points, respectively. These results establish Cognitive Core as a balanced foundation that combines physical-world cognition with the multimodal reasoning, tool use, and instruction-following capabilities required for experience evolution.

7.2 Evaluation of Action Model

We evaluate ME-Brain on history-dependent manipulation, general manipulation capabilities, and real-world robot deployment. RoboMME tests whether the model can use past visual evidence to guide current actions. RoboDojo evaluates generalization, precision, long-horizon execution, memory, and

Table 3. Success rates on RoboMME (%). AVG is the equally weighted mean across the four suites.

Method	Counting	Permanence	Reference	Imitation	AVG
$\pi_{0.5}$	28.0	19.5	16.0	9.0	18.12
FrameSampling + Modulation	66.0	24.0	36.5	52.0	44.62
MemER	48.5	51.5	42.0	25.0	41.75
ME-Brain	70.0	27.5	39.0	55.0	47.88

Table 4. Results on the RoboDojo simulation benchmark. Comparison results are taken from the [official RoboDojo leaderboard](#). Score ranges from 0 to 100, and SR is reported in percent. Generalization aggregates Gen-Std and Gen-Rand, and AVERAGE equally weights the five capability dimensions. Best results are boldfaced, and the ME-Brain rows are shaded.

Model	Metric	AVERAGE	Generalization	Precision	Long-Horizon	Memory	Open
GalaxeaVLA (G0.5)	Score	20.23	18.46	28.25	44.12	8.61	1.73
	SR (%)	14.88	12.83	20.42	32.25	7.33	1.58
Xiaomi-Robotics-1	Score	20.07	23.54	26.69	38.39	7.81	3.94
	SR (%)	13.93	17.00	18.83	23.67	6.56	3.58
OpenWAM- α	Score	17.18	20.71	18.45	34.93	10.41	1.41
	SR (%)	11.92	14.83	9.25	25.33	9.11	1.08
$\pi_{0.5}$	Score	11.41	13.38	12.40	23.54	5.78	1.98
	SR (%)	6.91	8.17	5.50	14.67	4.56	1.67
ME-Brain	Score	21.51	21.95	22.04	34.55	25.57	3.45
	SR (%)	16.03	16.00	12.75	24.00	24.67	2.75

open-vocabulary instruction following. Experiments on a physical Piper dual-arm robot further assess execution performance during real-world interaction.

7.2.1 Memory Benchmark

We first evaluate the role of historical information in action generation on RoboMME [38]. The benchmark contains 16 tasks organized into four suites according to their memory requirements: Counting, Permanence, Reference, and Imitation, with four tasks per suite. Table 3 reports the arithmetic mean of task success rates within each suite and the equally weighted mean across the four suites. We compare ME-Brain with $\pi_{0.5}$, FrameSampling + Modulation, and MemER [13].

ME-Brain achieves an overall success rate of 47.88%, improving over the in-house $\pi_{0.5}$ baseline, FrameSampling + Modulation, and MemER by approximately 29.8, 3.3, and 6.1 percentage points, respectively. Compared with the baseline that combines historical-frame sampling and modulation, ME-Brain improves performance on all four suites. In particular, it reaches 70.0% on Counting and 55.0% on Imitation, indicating that event counts and historical action patterns can inform current manipulation decisions. The consistent gains across these memory requirements suggest that action-conditioned history retrieval and event-driven action generation work effectively together.

7.2.2 Action Benchmark

We further evaluate general manipulation capabilities on the RoboDojo simulation benchmark [39]. The evaluation covers five capability dimensions on the ARX X5 dual-arm platform. Generalization is tested in both standard scenes (Gen-Std) and randomized scenes (Gen-Rand). Each of the 54 task configurations is evaluated over 24 trials, yielding 1296 valid trials in total. We report both terminal success rate (SR) and Score, a measure of task completion progress. Following the benchmark’s aggregation scheme, Generalization combines Gen-Std and Gen-Rand, while AVERAGE equally weights Generalization, Precision, Long-Horizon, Memory, and Open.

As shown in Table 4, ME-Brain achieves an average Score of 21.51 and an average SR of 16.03%, exceeding the official $\pi_{0.5}$ baseline by 10.10 points and 9.12 percentage points, respectively. ME-Brain improves both metrics across all five capability dimensions. The largest gain appears in Memory, where Score rises from 5.78 to 25.57 and SR from 4.56% to 24.67%. It also improves Long-Horizon Score by 11.01 points and SR by 9.33 percentage points, while Precision gains 9.64 points in Score and 7.25 percentage points in SR. Among the five systems in Table 4, ME-Brain ranks first overall in both Score and SR, leading the second-best system by 1.28 points and 1.15 percentage points, respectively. It also achieves top-two performance on Generalization, Memory, and Open for both metrics. In particular, ME-Brain ranks first on Memory with a Score of 25.57 and an SR of 24.67%, exceeding the second-best results by 15.16 points and 15.56 percentage points. These results demonstrate that event-target guidance, local future modeling, and action-conditioned memory modulation form a balanced policy with distinctive strength in memory-dependent control.

7.3 Real-World Robot Evaluation

We introduce ME-RealBench to systematically evaluate ME-Brain’s task execution capabilities in real-world physical environments. Deployed on a Piper dual-arm robot, the benchmark comprises six tasks: stacking bowls, stacking blocks and covering them with a cup, standing a bottle upright, placing objects in a basket, inserting pens into a pen holder, and plugging in a charger. These tasks cover key capabilities including dual-arm coordination, fine-grained manipulation, and multi-step task execution. Each task is evaluated over 10 independent trials. Success Rate (SR) is computed from the binary terminal success outcome of each trial. Partially completed trajectories receive a Score between 0 and 100 based on RoboDojo’s task-specific scoring levels, capturing task progress even when the task is not fully completed. Three evaluators independently score each trial under a double-blind protocol. Their ratings are averaged, followed by averaging over trials within each task and then equally over tasks. TimeExpend is measured from the policy’s first action output until task success, failure, timeout, or a safety stop, and is averaged over successful trials only.

ME-Brain achieves a mean SR of 66.7% and a mean Score of 69.5 across the six tasks, the highest values among the four evaluated models (Table 5). Relative to DM0.5, these metrics improve by 11.7 percentage points and 12.8 points, respectively; relative to $\pi_{0.5}$, the improvements are 16.7 percentage points and 17.3 points. ME-Brain obtains a non-zero Score on every task and succeeds in all 10 bowl-stacking trials. On the more challenging block-covering task, it reaches a 50% SR, 40 percentage points above DM0.5. On the pen-holder task, which requires repeated grasping and placing, its Score reaches 72, improving over the $\pi_{0.5}$ Score of 8 by 64 points. Together, the simulation and real-world results demonstrate effective execution across diverse manipulation tasks for the interaction-event-centric policy.

Table 5. Evaluation on the six-task ME-RealBench using a Piper dual-arm robot. (a) Overall results: SR and Score are equally weighted means across tasks; TimeExpend is the mean execution time over successful trials only. (b) Per-task results: SR is reported in percent and Score ranges from 0 to 100, with 10 trials per task. Higher SR and Score and lower TimeExpend are better. Best values for each metric are boldfaced, including ties.

(a) Overall performance							
Model		SR (%) $\uparrow$		Score $\uparrow$		TimeExpend (s) $\downarrow$	
ME-Brain		66.7		69.5		75.5	
DM0.5		55.0		56.7		89.1	
$\pi_{0.5}$		50.0		52.2		67.5	
xr-1		23.3		29.8		77.4	

(b) Per-task performance							
Model	Metric	Stack bowls	Cover blocks	Stand bottle	Place in basket	Fill pen holder	Plug in charger
ME-Brain	SR (%)	100	50	90	90	60	10
	Score	100	50	95	90	72	10
DM0.5	SR (%)	100	10	80	60	80	0
	Score	100	10	84	60	86	0
$\pi_{0.5}$	SR (%)	100	0	90	100	0	10
	Score	100	0	95	100	8	10
xr-1	SR (%)	70	0	50	20	0	0
	Score	73	15	61	20	10	0

8 Qualitative Examples

This section presents six real-world deployment cases that qualitatively illustrate how the three core modules of ME-Brain work together. Spanning capabilities from basic manipulation to memory-guided reasoning, these cases show how the system integrates Evolvable Memory, Cognitive Core, and Action Model into coherent task execution in physical environments.

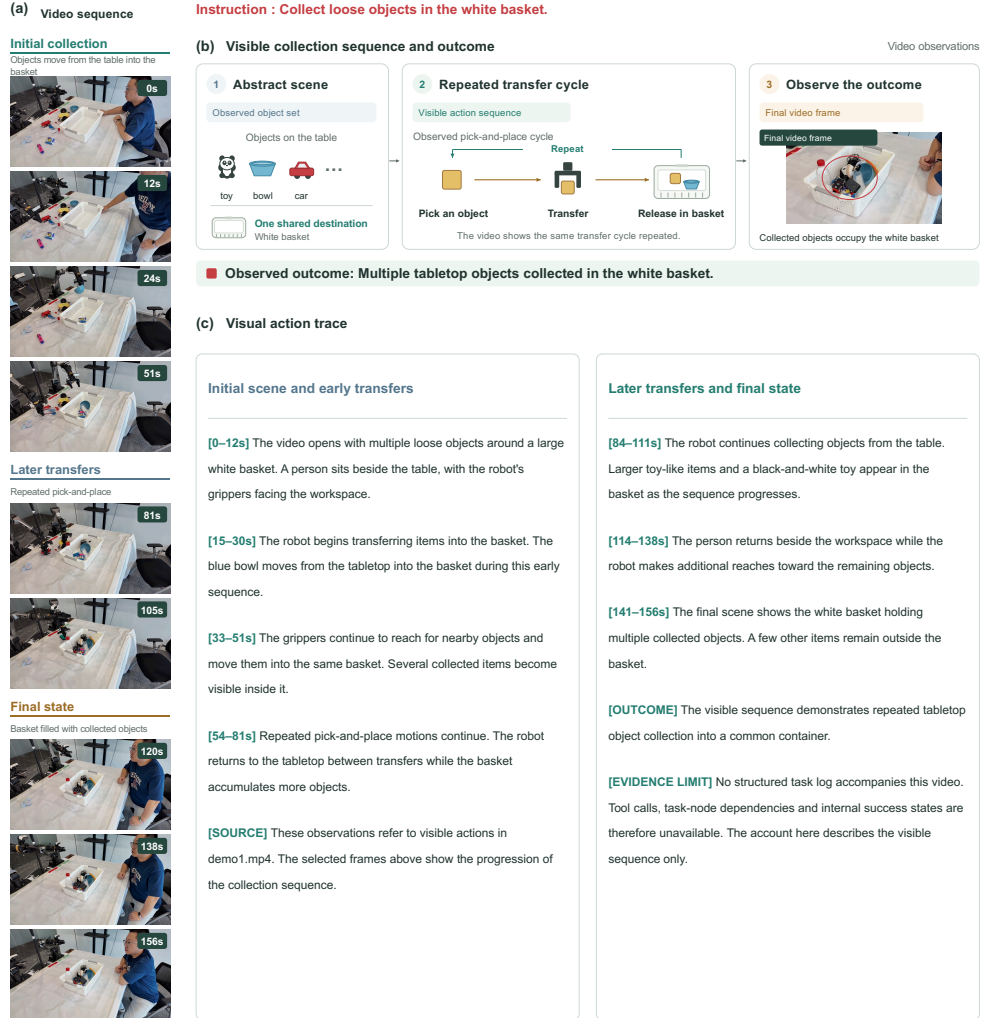


Figure 5. Multi-object tabletop collection

The first case illustrates ME-Brain’s ability to sustain execution over repeated manipulation cycles. The robot repeatedly grasps, transfers, and releases objects scattered across the tabletop into the same white basket. As the target objects and their positions change throughout execution, the robot returns to the tabletop between transfers to locate and grasp additional objects, progressively completing the collection process. The key frames show the initial scene, intermediate object transfers, and the growing number of objects in the basket, illustrating the system’s ability to maintain a repeated action pattern during continuous manipulation. This analysis is based primarily on actions visible in the video and documents the ongoing collection of multiple objects.

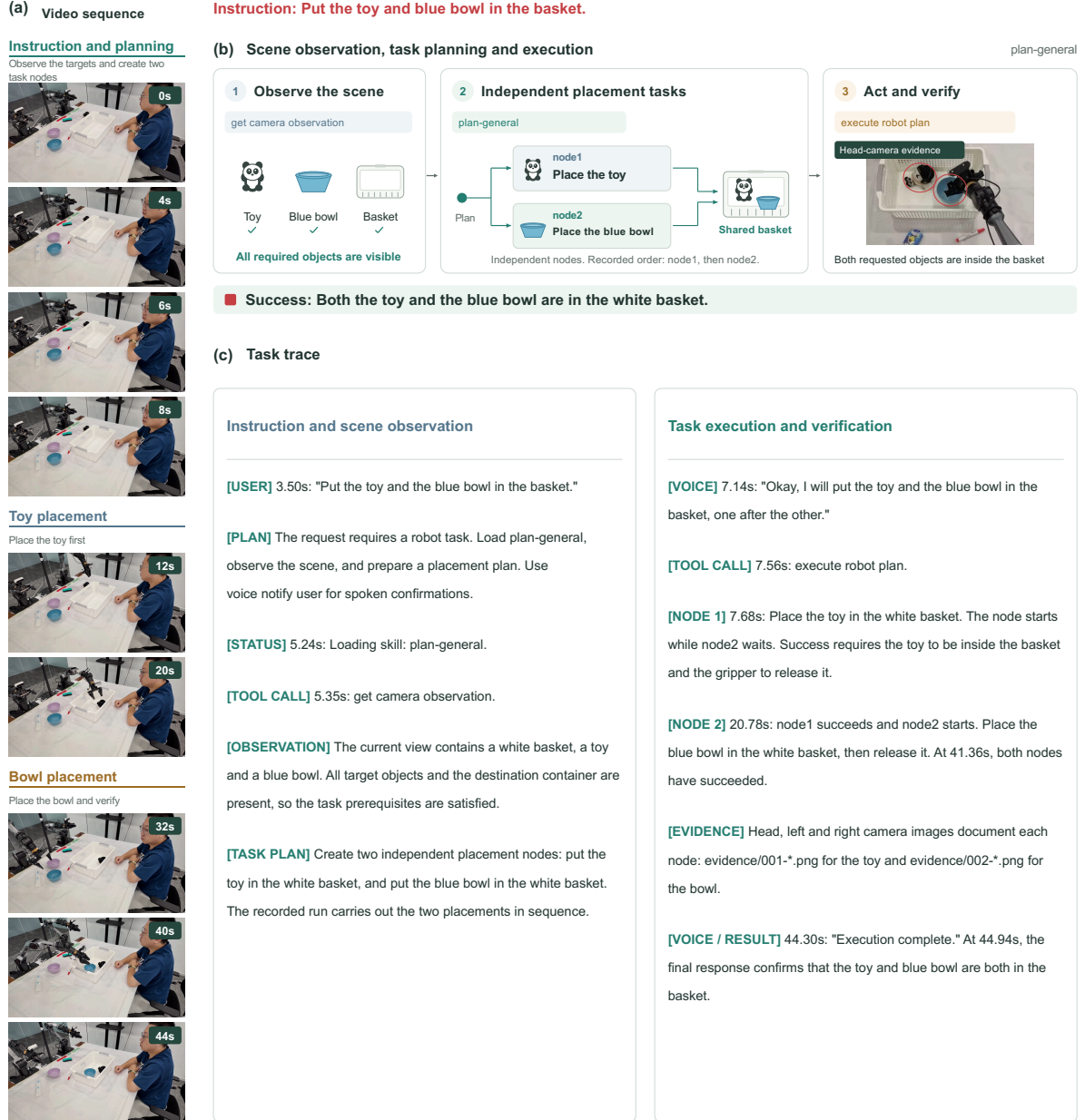


Figure 6. Instruction-guided object placement

The second case illustrates ME-Brain’s ability to translate a language instruction with multiple goals into structured subtasks and execute them sequentially. Given the instruction “Put the toy and the blue bowl in the basket,” the system first uses current visual observations to locate both target objects and the white basket, then creates two task nodes for placing the toy and the blue bowl. The two subgoals are logically independent. In this run, the system first places the toy in the basket and releases the gripper, then starts the blue-bowl placement, ultimately completing both subgoals. The task trace records scene observation, task planning, node execution, and completion feedback, illustrating how ME-Brain extracts target objects and their placement relations from language and translates them into a sequence of robot actions.

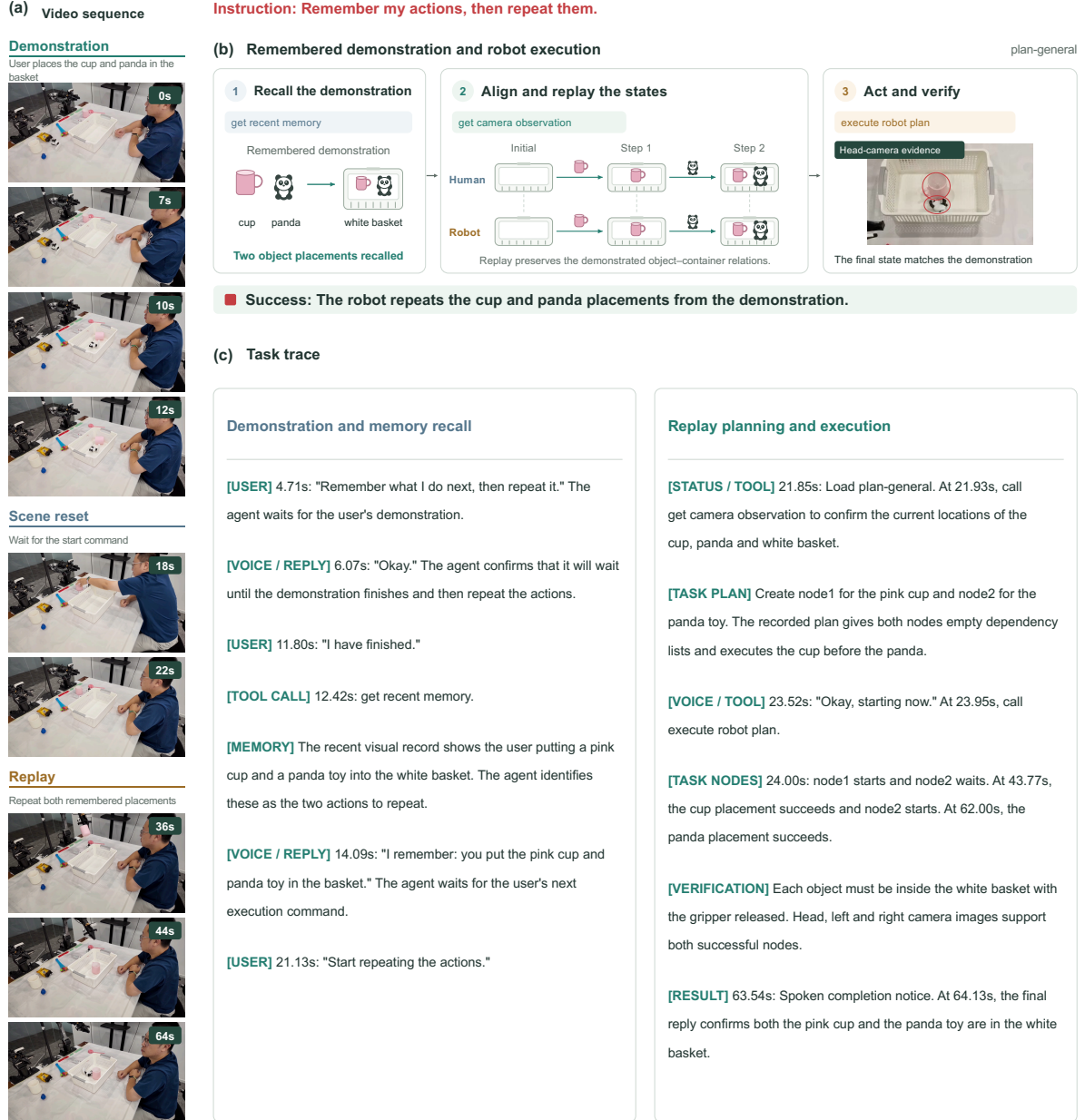


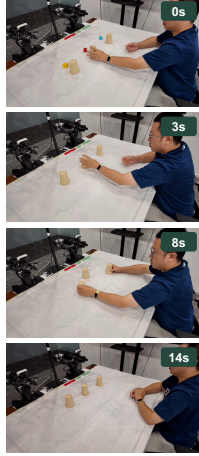
Figure 7. Visual memory-guided demonstration replay

The third case illustrates ME-Brain’s ability to reproduce a human manipulation demonstration using recent visual memory. After receiving the instruction “Remember my actions, then repeat them,” the system waits for the user to complete the demonstration. The user places a pink cup and a panda toy into the white basket and announces that the demonstration is complete. The system then retrieves recent visual memory to identify and confirm the two placement actions to be repeated. After the scene is reset and the objects are returned to the tabletop, the user gives the start command. The system confirms the current locations of the objects and basket, then places the pink cup and the panda toy into the basket in sequence. This case reproduces both the object–container relations and the placement order in the demonstration, illustrating how recent visual memory preserves demonstration content and supports task execution by combining historical information with current scene observations.

(a) Video sequence and user instruction

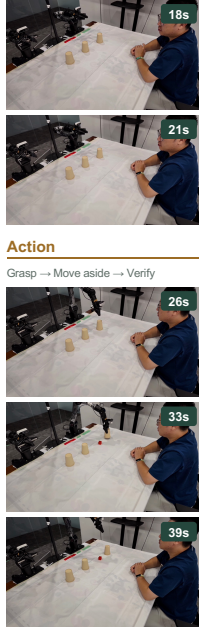
Memory

Cubes visible → Cubes covered → Cup swaps



Current buffer

Target occluded

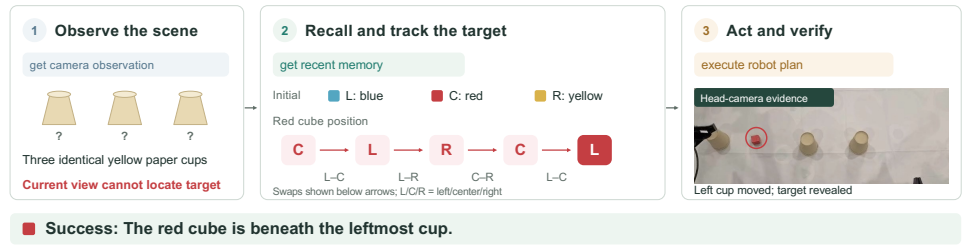


Action

Grasp → Move aside → Verify

Instruction: Find the red cube.

(b) Memory-grounded localization and action



(c) Complete task trace

Planning and current observation

Answer: The user asked me to find the red cube. This robotic task requires first loading the relevant skill to determine the execution procedure. Among the available skills, I should use plan-general for task planning. Next, I need to: 1. Load plan-general; 2. Observe the scene following the skill's procedure; 3. Locate and find the red cube. I will first load plan-general.

[STATUS] Loading skill: plan-general

The plan-general procedure requires observing the current scene first. I should therefore call get camera observation to obtain current visual observations for subsequent target identification and localization.

[TOOL CALL] get camera observation

[TOOL RESPONSE] Only three identical yellow paper cups are on the table; the red cube requested by the user was not found.

The current view alone cannot locate the target. I therefore need to call get recent memory to retrieve recent visual observation records and determine whether the red cube appeared earlier and where it might be.

[TOOL CALL] get recent memory

Memory-based localization and execution

[TOOL RESPONSE] Initially, the blue cube was beneath the left paper cup, the red cube beneath the center paper cup, and the yellow cube beneath the right paper cup. The user then covered the cubes with the three paper cups. Next, the user swapped the cups in this order: left-center, left-right, center-right, and left-center.

Tracking the changes in cup positions shows that the red cube ends beneath the left paper cup. With the target location now determined, a task node can be created. I will create the task node: "Grasp the left paper cup and move it to an adjacent position."

[TOOL CALL] voice notify user {"voiceText": "Okay, the red cube ends up beneath the leftmost cup."}

[TOOL CALL] execute robot plan

[TASK NODES] node1 "Grasp the left paper cup and move it to an adjacent position" Status: In progress → Succeeded; Evidence: evidence/001-head.png, evidence/001-left.png, evidence/001-right.png

[TOOL CALL] voice notify user {"voiceText": "Execution complete."}

The red cube beneath the leftmost cup was successfully found. Task execution is complete.

Figure 8. Visual memory-guided localization and action

The fourth case illustrates ME-Brain's ability to locate an occluded object using a sequence of past events. Given the instruction "Find the red cube," the current observation shows only three identical yellow paper cups, making it impossible to determine the target's location directly. By retrieving recent visual memory, the system establishes that the red cube was initially beneath the center cup and reconstructs four subsequent cup swaps: left-center, left-right, center-right, and left-center. Tracking these position changes step by step, the system infers that the red cube is now beneath the leftmost cup and commands the robot to move that cup aside, revealing the target. Whereas the third case reproduces demonstrated placement relations, this case requires the system to integrate state changes across multiple time steps, infer the current location of an invisible target from past events, and verify the inference through physical action.

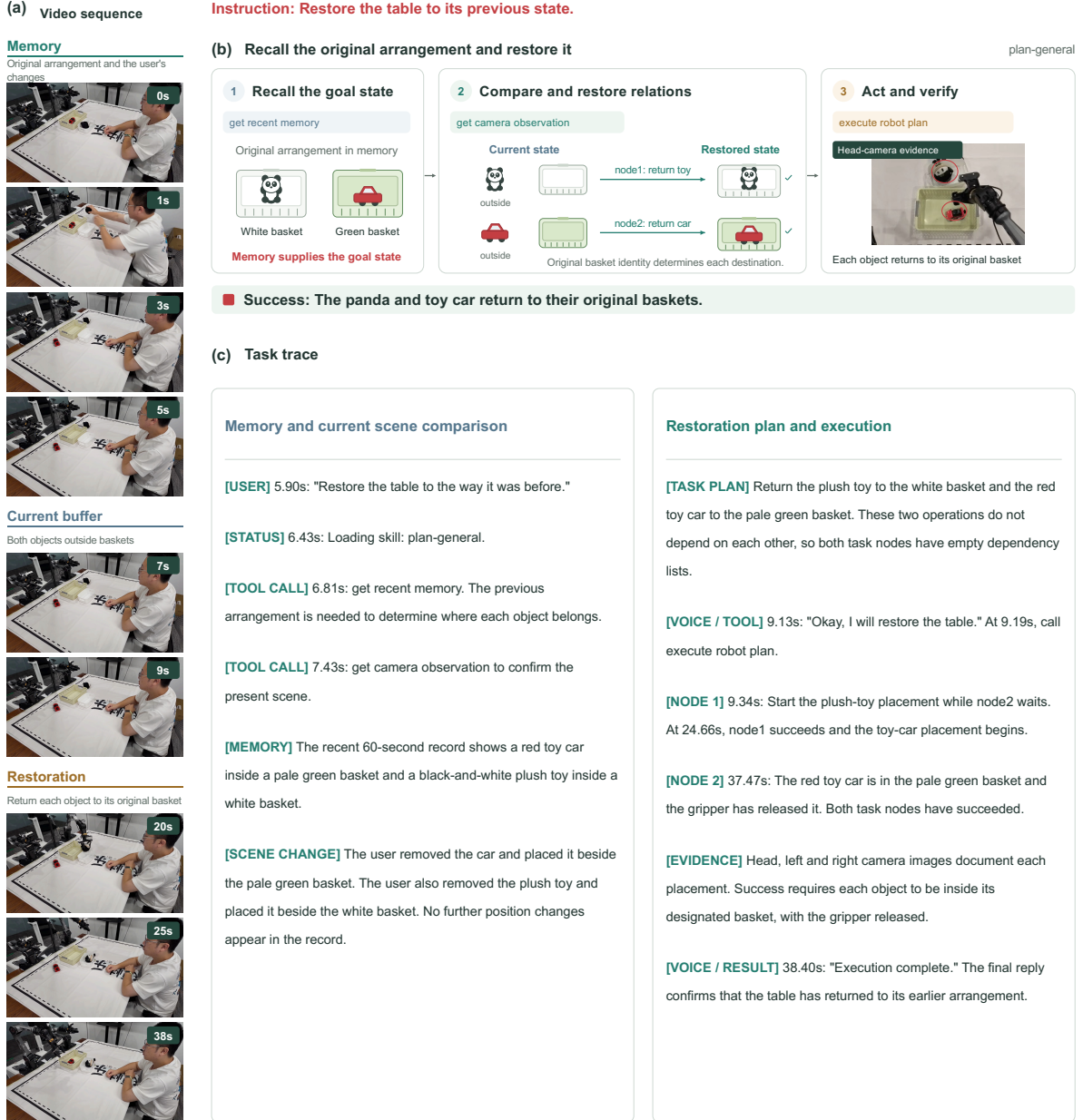


Figure 9. Visual memory-guided scene restoration

The fifth case illustrates ME-Brain’s ability to recover a task’s goal state from a past scene. After removing objects from their original containers, the user instructs the system to “Restore the table to its previous state” without specifying a destination for each object. The system first retrieves recent visual memory to establish that the panda toy was originally in the white basket and the red toy car in the pale green basket. It then combines this information with current observations to determine the required restoration actions. The robot returns the panda toy to the white basket, followed by the red toy car to the pale green basket, restoring the original object–container relations. Compared with demonstration replay and occluded-object localization, this case emphasizes retrieving and restoring a historical scene state: the system must recover goal relations left unspecified by the current instruction from memory and use them to generate a concrete execution plan.

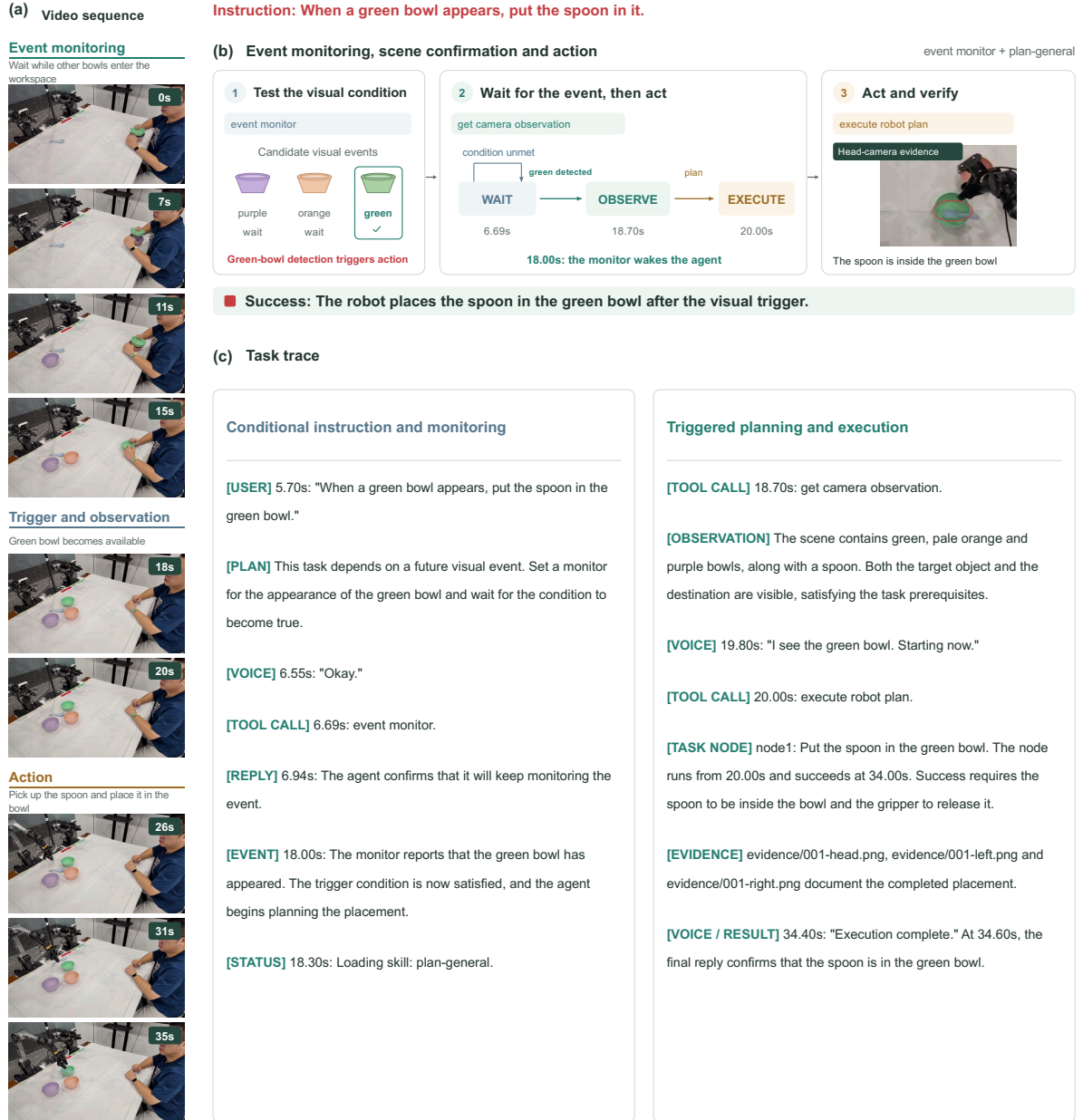


Figure 10. Visual event-triggered object placement

The sixth case illustrates ME-Brain’s ability to interpret conditional language instructions and execute actions in response to an event. Given the instruction “When a green bowl appears, put the spoon in it,” the system sets the appearance of a green bowl as the condition for visual event monitoring and enters a waiting state. It continues waiting as a purple bowl and an orange bowl appear in succession, since neither satisfies the target condition. Once the monitor reports the appearance of the green bowl, the system obtains current visual observations, confirms the locations of the spoon and the green bowl, and plans and executes the placement task, finally placing the spoon in the bowl and releasing the gripper. In this case, the user specifies the task in advance, while a subsequent environmental event determines when execution begins. The case illustrates how ME-Brain converts a conditional language instruction into a monitoring condition and, once that condition is met, connects scene confirmation, task planning, and robot execution.

9 Conclusion and Future Work

9.1 Conclusion

This work addresses the limitation that embodied models largely remain fixed after training and cannot continuously learn from ongoing interaction. We propose *ME-Brain*, a self-evolving embodied system that decomposes post-deployment self-evolution into three interconnected problems: experience acquisition, experience evolution, and action execution. These problems are respectively handled by *Evolvable Memory*, *Cognitive Core*, and *Action Model*, forming a self-driven loop of “execution–acquisition–evolution–execution” that supports autonomous growth without model retraining. Evolvable Memory constructs an external memory graph through three-level hierarchical storage and a progressive “record–summarize–abstract” process, enabling experience to accumulate and transfer across tasks. Cognitive Core unifies embodied cognition with multimodal agent capabilities, grounding experience in the physical world and consolidating reusable knowledge into skills. Focus-VLWA in the Action Model turns world modeling into local dynamics supervision that directly supports action learning. Together, the three modules allow execution to produce experience and accumulated experience to improve subsequent actions, shifting the embodied system from a “train-and-freeze” paradigm toward a “deploy-and-evolve” paradigm. Through hardware–software co-optimization with M100, all three ME-Brain modules can also run independently on the edge chip, providing an engineering foundation for scalable deployment in real physical environments. This framework offers a feasible path toward embodied systems capable of long-term autonomous growth in open physical environments.

9.2 Future Work

Future work will focus on the lifelong operation of the memory module and the evaluation of long-term embodied intelligence. For the memory system, we will further investigate memory compression, redundancy removal, and dynamic activity monitoring, using task relevance, access frequency, and temporal validity to identify and remove duplicated or low-value nodes and to consolidate information that no longer requires fine-grained representations. This will allow the system to preserve critical experiences and necessary associations while controlling storage and retrieval costs, preventing the continuous accumulation of historical data from degrading retrieval and maintenance efficiency and enabling memory to evolve throughout long-term interaction. Building on this capability, we will further explore secure on-device deployment of lifelong memory under resource-constrained settings, where user-related observations, interactions, and task experiences can be maintained and retrieved locally, together with improved data protection and memory management mechanisms. Through long-term interaction, the system can gradually accumulate knowledge of user preferences, behavioral patterns, and needs, and continuously update this knowledge as the user’s routines and living context change, ultimately transforming long-term memory into a persistent and personalized service capability that enables embodied robots to better understand users and adapt to their evolving needs. In terms of evaluation, we will further conduct controlled comparisons and ablation studies covering long-term operation, environmental changes, and skill reuse, systematically examining the effects of memory and skill updates on task success rate, failure recovery capability, and computational and storage overhead.

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10 Contributions and Acknowledgments

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Acknowledgments

Xueyang Zhang, Mofan Zhou, Bing Zhang, Yuying Chen, Shengyu Yao, Chang Ren, Chaoqun Du, Ming Li, Danlu Dong, Mingcui Wang, Jiao Deng, Yue Ma, Lingling Lu, Depeng Xu, Yimeng Li, Chen Liu, Liming Liu, Chunrui Wang